

Effects of nano- and micro-limestone addition on early-age properties of ultra-high-performance concrete

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Abstract In this study, the effects of micro- and nano- CaCO_3 addition on the early-age properties of ultra-high-performance concrete (UHPC) cured at simulated cold and normal field conditions were investigated. The micro- CaCO_3 was added at rates of 0, 2.5, 5, 10 and 15 %, while the nano- CaCO_3 was added at rates of 0, 2.5, and 5 %, both as partial volume replacement for cement. Results indicate that micro- CaCO_3 acted mainly as an inert filler, creating a denser microstructure and increasing the effective w/c ratio. In addition, nano- CaCO_3 accelerated the cement hydration process through nucleation, and also acted as an effective filling material. Mixtures combining both micro- and nano- CaCO_3 resulted in similar or enhanced mechanical properties compared to that of the control, while achieving cement replacement levels up to 20 %. Thus, through the use of micro- and nano- CaCO_3 , more environmentally friendly UHPC can be produced by reducing its cement factor, while achieving enhanced engineering properties.

Keywords Nano- CaCO_3 · Micro- CaCO_3 · Early-age · Green concrete · Drying shrinkage

1 Introduction

Concrete is the most widely used construction material in the world. Over 10 billion tons are produced annually [1], with a serious environmental and economic impact. Just one ton of portland cement produced emits ~ 1 ton of carbon dioxide (CO_2) into the atmosphere [1]. In addition, it consumes substantial natural resources and energy. In order to achieve “greener” concrete, cement can be partially replaced by inert materials such as calcium carbonate (CaCO_3), or by supplementary cementitious materials (SCM's) such as fly ash, silica fume, and ground granulated blast furnace slag [2].

The use of CaCO_3 is an attractive alternative. It can be found in limestone, marble and chalk, or produced artificially by combining calcium with carbon dioxide. Variations in temperature and pressure during production are able to produce CaCO_3 in the form of calcite, aragonite, and vaterite [3]. Previous studies focused primarily on using micro calcite because it is the most stable form of CaCO_3 . This compound is known to react chemically, accelerate the hydration process, and in turn, increase the early-age strength of conventional cementitious materials [4].

With recent enormous developments in concrete technology, new generations of concrete are being produced, such as very high-strength concrete (VHSC) and ultra-high-performance concrete (UHPC). These new concrete types are characterized by very low water-to-cement (w/c) ratios [5]. Hence, high percentages of the added cement will not hydrate and will

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merely act as filler ($\sim 42\%$) [6]. Therefore, replacing cement with a non-costly filler material will reap economic and environmental benefits. It has been suggested [7] that cement can be partially replaced with limestone (consists mainly of CaCO_3), with little or no effect on the hydration process and compressive strength. Limestone has been commonly used in European cements, but remains limited by cement standards in North America [7]. The Canadian Standards Association has been discussing protocols for the use of limestone additions in cement-based materials and the allowable amount limits for various applications and environments [8].

On the other hand, since the invention of nano-materials, the concrete industry has made significant investments into exploring the effects of nano-materials in concrete and their potential applications [9–12]. Many nano-additives create superior results to that of their micro counterparts. For example, a previous study showed that the addition of nano-silica ($\approx 40\text{ nm}$ ($1.57\text{ }\mu\text{in}$)) had improved the early-age compressive strength of mortar by approximately twice that induced by micro-silica fume (SF) addition ($0.1\text{ }\mu\text{m}$ ($3.94\text{ }\mu\text{in}$)) [13]. The advantage of nano- CaCO_3 compared to other nano-materials is that it has a relatively reasonable cost [14]. Limited research has investigated the effect of nano- CaCO_3 addition in cement-based materials, its benefits and potential applications.

This paper investigates the effects of incorporating micro- and nano- CaCO_3 , both individually and combined, as partial replacement for cement on the early-age properties of UHPC. This will emphasize the difference in the cement hydration accelerating effect induced by nano- CaCO_3 as opposed to the reagent grade micro- CaCO_3 . To the best of the authors' knowledge, no previous research has been conducted on the effects of using combined micro- and nano-sized CaCO_3 in UHPC. Hence, this work should contribute to the body of knowledge on the influence of different sizes of limestone particles, and their combinations on the early-age properties of UHPC.

2 Research significance

The use of micro- and nano-sized CaCO_3 additives, single and combined, as partial replacement for cement has not yet been studied in depth. In UHPC, which is characterized by very low w/c, a high portion

of the portland cement remains unhydrated, and thus could be partially replaced by more economical and environmentally friendly materials. From what is currently known about nano-materials, the ultrafine size offers significant beneficial properties compared to coarser additives. Thus, in this paper the effects of micro- and nano- CaCO_3 as partial replacement for cement, and their benefits in UHPC have been investigated. The study provides an innovative solution for reducing the cement factor in UHPC, thus achieving “greener” concrete.

3 Experimental program

This experimental program aims at gaining an understanding of the mechanisms involved in using micro- and nano- CaCO_3 in UHPC. In this study, monitoring of the hydration process (degree of hydration, heat of hydration, setting time), characterizing of the mechanical properties and microstructural development have been carried out on UHPC mixtures incorporating micro- and nano- CaCO_3 in order to evaluate its efficiency as a partial replacement for portland cement.

3.1 Materials and mixture proportions

An ordinary portland cement (OPC) and SF were used as binders. The chemical and physical properties of the various binders are listed in Table 1. According to the suggestions of [15], coarse aggregates were not used in UHPC. Quartz sand having a particle size in the range of $0.1\text{--}0.8\text{ mm}$ ($0.004\text{--}0.031\text{ in}$) was used instead. A polycarboxylate high-range water-reducing admixture (HRWRA) was added at a rate of 3% by mass of cement. Micro- CaCO_3 having $3\text{ }\mu\text{m}$ average particle size, and a nano-particle sized ($15\text{--}40\text{ nm}$ ($0.59\text{--}1.57\text{ }\mu\text{in}$)) CaCO_3 were added as a white powder at rates of $0, 2.5, 5, 10$, and 15% for micro- CaCO_3 and $0, 2.5$, and 5% for nano- CaCO_3 as partial volume replacement for cement. Figure 1a, b shows the particle-size distribution for the used micro- CaCO_3 and an SEM micrograph of the nano- CaCO_3 powder. The particle size distribution shows slightly higher sizes due to agglomeration of the fine particles [16]. The mixtures are labeled with respect to their dosage: #N-#M; “N” indicating nano-particles, “M” indicating micro-particles, and “#” representing the corresponding



Table 1 Chemical and physical properties of cement and supplementary cementitious materials

	OPC	Silica fume	Limestone
SiO ₂ (%)	19.8	94.0	–
CaO (%)	63.2	0.4	–
Al ₂ O ₃ (%)	5.0	0.1	–
Fe ₂ O ₃ (%)	2.4	0.1	–
MgO (%)	3.3	0.4	<0.45
K ₂ O (%)	1.2	0.9	–
SO ₃ (%)	3.0	1.3	–
Na ₂ O (%)	0.1	0.1	–
TiO ₂ (%)	0.3	0.3	–
CaCO ₃ (%)	–	–	99.0
Loss on ignition (%)	2.5	4.7	–
Specific surface area (m ² /kg)	410	19,530	3,200
Specific gravity	3.17	2.12	2.70
C ₃ S	61	–	–
C ₂ S	11	–	–
C ₃ A	9	–	–
C ₄ AF	7	–	–

dosage. For example, 2.5 N–5 M represents a mixture with 2.5 % nano-CaCO₃ and 5 % micro-CaCO₃, together partially replacing 7.5 % of cement by volume. Before mixing the UHPC, nano-CaCO₃ particles were dispersed in the mixing water using the ultrasonic dispersion method [17]. Water from the HRWRA was included in the specified w/c. The selected composition of the control mixture, which is a well-known class of UHPC without coarse aggregate [18, 19], and the

characteristics of the tested mixtures are shown in Tables 2 and 3, respectively.

3.2 Test methods and specimens preparation

The workability of each mixture was evaluated based on the flow index (F), which is defined as follows (Eq. 1):

$$F(\%) = \frac{R_{25} - R_0}{R_0} \times 100 \quad (1)$$

where R_{25} is the radius of the mortar pile after the 25th drop and R_0 is the initial radius of the mortar pile according to ASTM C 1437 (standard test method for flow of hydraulic cement mortar) [20].

Compressive strength testing was conducted on 50 mm (2 in.) UHPC cubes at the ages of 6, 8, 10, and 12 h, and 1, 3, 7, and 28 days using a 200-ton (441 kip) compression testing machine (innovative instruments). Specimens were moist cured under burlap for the first 24 h, and then submerged in lime-saturated water for the remainder of the curing period. Two curing regimes, namely at 10 and 20 ± 1 °C (50 and 68 °F), were conducted in a walk-in environmental chamber to simulate cold and normal site conditions. The time of setting was measured on three replicate paste specimens using a Vicat needle according to ASTM C191 (standard test method for time of setting of hydraulic cement by Vicat needle) [21].

Semi-adiabatic calorimetry studies were conducted on UHPC specimens during the first 2 days of hydration using a custom-built experimental setup.

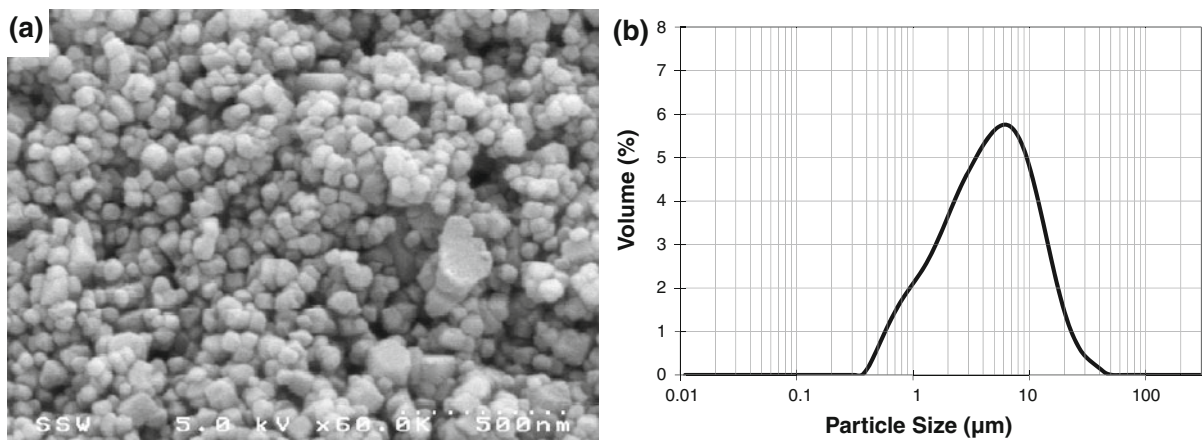


Fig. 1 Particle size distribution of CaCO₃ powders: **a** SEM micrograph for nano-CaCO₃, **b** laser diffraction spectrometry for 3 μm-CaCO₃

Table 2 Composition of control mixture

Material	(Mass/ cement mass)
Cement	1.00
Silica fume	0.30
Quartz sand (0.1–0.5 mm)	0.43
Quartz sand (0.3–0.8 mm)	1.53
Water	0.25
HRWRA	0.03

The UHPC was prepared and cast into a prismatic mold [60 × 100 × 250 mm (2.5 × 4 × 10 in.)]. The mold was immediately placed in a micro-porous insulation box. Three type-T thermocouples were inserted into the center of the concrete volume along its length to monitor its temperature. Replicate specimens indicated a standard deviation of 1.8 °C (35 °F) for the maximum specimen temperature.

Thermo-gravimetric analysis (TGA) combined with derivative thermo-gravimetric (DTG) was used to determine the evolution of the bound water (BW) content during hydration. This indirect method has been commonly used e.g. [22, 23] to quantify the

degree of hydration. Since only one binder composition was used, a linear correlation between the amount of BW and the degree of hydration was assumed, in agreement with previous studies [22, 23]. The sample preparation and test procedure are presented in greater detail elsewhere [22]. These tests were conducted using a simultaneous DSC-TGA (Model SDT 2960).

For the drying shrinkage measurements, prismatic specimens 25 × 25 × 285 mm (1 × 1 × 11 in.) were prepared according to ASTM C 157 (standard test method for length change of hardened hydraulic-cement mortar and concrete) [24]. Immediately after demolding (i.e. at age 24 h), the initial lengths of specimens were measured, then specimens were moved inside the walk-in environmental chamber. Drying was conducted at 10 and 20 ± 1 °C (50 and 68 °F) inside the environmental chamber with a relative humidity of 40 %. The unrestrained one-dimensional deformations have been measured using a digital comparator provided by a dial gauge with an accuracy of 10 µm/m. Small prismatic cross-section [25 × 25 mm (1 × 1 in.)] specimens were chosen to reduce the moisture gradients effect induced by drying [23] and to assure quick dissipation of the hydration heat [25, 26].

Table 3 Tested mixtures

Mixture	Nano-CaCO ₃ content (%) ^a	Micro-CaCO ₃ content (%) ^a	Total cement replacement (%) ^a	Compressive strength (% of control)			
				24 h		28 days	
				10 °C (50 °F)	20 °C (68 °F)	10 °C (50 °F)	20 °C (68 °F)
Control	0.00	0.00	0.00	100.0	100.0	100.0	100.0
0 N–2.5 M	0.00	2.50	2.50	153.3	122.2	98.7	105.4
0 N–5 M	0.00	5.00	5.00	143.7	103.5	101.6	103.1
0 N–10 M	0.00	10.00	10.00	126.7	101.9	96.9	100.3
0 N–15 M	0.00	15.00	15.00	107.0	97.0	94.4	97.3
2.5 N–0 M	2.50	0.00	2.50	132.5	161.9	113.7	93.8
2.5 N–2.5 M	2.50	2.50	5.00	162.9	121.6	95.7	97.4
2.5 N–5 M	2.50	5.00	7.50	176.8	108.4	98.0	82.3
2.5 N–10 M	2.50	10.00	12.50	159.9	92.0	97.2	88.9
2.5 N–15 M	2.50	15.00	17.50	162.3	81.6	101.6	85.0
5 N–0 M	5.00	0.00	5.00	174.8	161.4	123.7	90.3
5 N–2.5 M	5.00	2.50	7.50	163.2	108.1	107.5	100.0
5 N–5 M	5.00	5.00	10.00	172.8	111.8	105.0	100.6
5 N–10 M	5.00	10.00	15.00	164.2	111.9	99.0	98.6
5 N–15 M	5.00	15.00	20.00	149.7	87.9	103.8	91.0

^a As a volume replacement of cement



Prismatic specimens $25 \times 25 \times 280$ mm ($1 \times 1 \times 11$ in.) were made for measuring mass loss for each mixture. Specimens were demolded at the time of starting drying shrinkage measurements (24 h after casting). Prisms were transferred to the walk-in environmental chamber after measuring the initial mass of each prism using a balance with an accuracy of 0.01 g (0.00035 oz.). Mass measurements were taken for all prisms along with measurements of shrinkage strains. Each mass loss test result in this study represents the average value obtained on three identical prisms (maximum standard deviation of 0.18 g (0.0063 oz)).

Particle sizes were obtained using a Mastersizer 2000 laser diffraction particle analyzer (Malvern Instruments) for micro-limestone, while scanning electron microscopy (Hitachi S-4500 Field Emission SEM) was used for illustrating the particles size at the nano level and examining the internal microstructure of the tested mixtures.

4 Results

4.1 Flowability

Figure 2 shows the changes in flowability for UHPC mixtures incorporating different contents of nano- CaCO_3 , while the added percentage of micro- CaCO_3 increased from 0 up to 15 %. Generally, all mixtures incorporating micro- and/or nano- CaCO_3 exhibited greater flowability compared to that of the control mixture without CaCO_3 addition. The flowability improved as nano- CaCO_3 was added to mixtures with constant amounts of micro- CaCO_3 . For instance in mixtures with 2.5 % micro- CaCO_3 , incorporating 2.5 and 5 % nano- CaCO_3 led to 14.5–40 % improvement in the flowability, respectively. Similarly, mixtures became more flowable as the micro- CaCO_3 content increased at constant nano- CaCO_3 addition rates. For example, mixtures with 2.5 % nano- CaCO_3 had an improvement in flowability of 14.7–25 % when micro- CaCO_3 addition was increased to 5–10 %, respectively. Therefore, greater cement replacement amounts achieved greater flowability. These trends can be attributed to the lubrication effect caused by the presence of fine CaCO_3 particles [27, 28]. Although particles with high surface area are being introduced into these mixtures, which would normally lower flowability due to higher water demand, the amount of

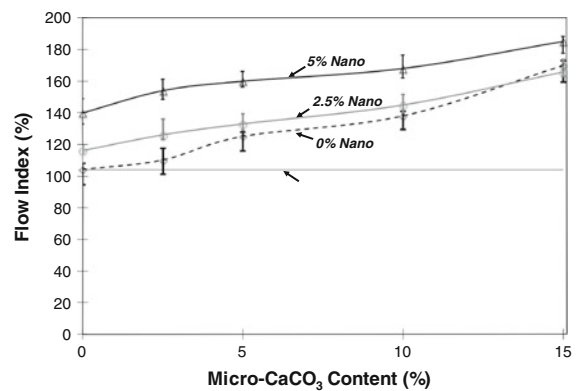


Fig. 2 Flow table results of UHPC mixtures incorporating various sizes and dosages of CaCO_3

dilution becomes greater, therefore increasing the effective w/c ratio. Furthermore, the CaCO_3 particles release entrapped water between coarser particles [27–30], therefore making more water available to aid in the flowability.

4.2 Setting time

During early-age, cement paste is plastic without a permanent internal structure. A few hours later, as hydration reactions proceed resulting in the formation of hydration products, a skeleton starts to develop and setting occurs [31]. Generally, any parameter that increases the hydration rate will shorten the setting time. Therefore, increasing the curing temperature from 10 °C (50 °F) to 20 °C (68 °F) will accelerate the formation of hydration products, thus leading to shorter setting time as shown in Fig. 3 [32]. For instance, mixtures incorporating 2.5 % micro- CaCO_3 exhibited a reduction of 11–32 % in the initial setting time and of 21–31 % in the final setting time, with respect to control mixture without CaCO_3 at 10 °C (50 °F) and 20 °C (68 °F), respectively.

In addition, increasing the number of nucleation sites will lead to a more rapid hydration rate [33]. Therefore, adding nano- CaCO_3 increased the hydration rate and shortened the setting time as shown in Fig. 3. At 10 °C (50 °F), higher reduction in setting time with nano- CaCO_3 addition was achieved compared to that at 20 °C (68 °F). For instance, increasing the nano- CaCO_3 addition from 0 to 5 % resulted in a reduction in the final setting time ranging from 38 to 52 % at 10 °C (50 °F) and from 33 to 35 % at 20 °C

(68 °F) as micro- CaCO_3 addition was increased from 2.5 to 15 %. This is likely due to the effect of the added nano- CaCO_3 on accelerating cement hydration reactions. Apparently, the relatively slower rate of hydration and microstructure development at the low temperature of 10 °C (50 °F) was compensated for by the presence of nano- CaCO_3 , leading to more hydration products and consequently a stronger microstructure [27]. Hence, the acceleration effect of the nano- CaCO_3 particles was more prominent at the colder temperature for these mixtures.

Regardless of the curing temperature, adding nano- CaCO_3 to UHPC mixtures at a constant micro- CaCO_3 content decreased the initial and final setting times compared to that of mixtures without nano- CaCO_3 . For instance, in mixtures with 10 % micro- CaCO_3 cured at 10 °C (50 °F), the final setting time was reduced by 39–48 % when the amount of nano- CaCO_3 increased to 2.5–5 %, respectively, compared to that

of mixtures without nano-particles (Fig. 3c). This is due to both an increased number of contact points between hydration products in the mixtures due to the presence of CaCO_3 [34], along with the accelerated hydration promoted by the nano-particles due to the nucleation effect [2], as can be observed in the heat of hydration results discussed later in this text.

At both curing temperatures, mixtures with similar nano- CaCO_3 content experienced a slight increase, if any, in the setting time when the amount of micro- CaCO_3 was increased. For instance at 20 °C (68 °F), the final setting time for mixtures incorporating 5 % nano- CaCO_3 increased by only 6 % when the added amount of micro- CaCO_3 increased from 0 to 15 % (Fig. 3d). This can be attributed to two compensating effects: dilution and filling. The cement dilution effect results in the formation of less hydration products within mixtures having high cement replacement levels, therefore causing slower setting times [35]. This is offset by the

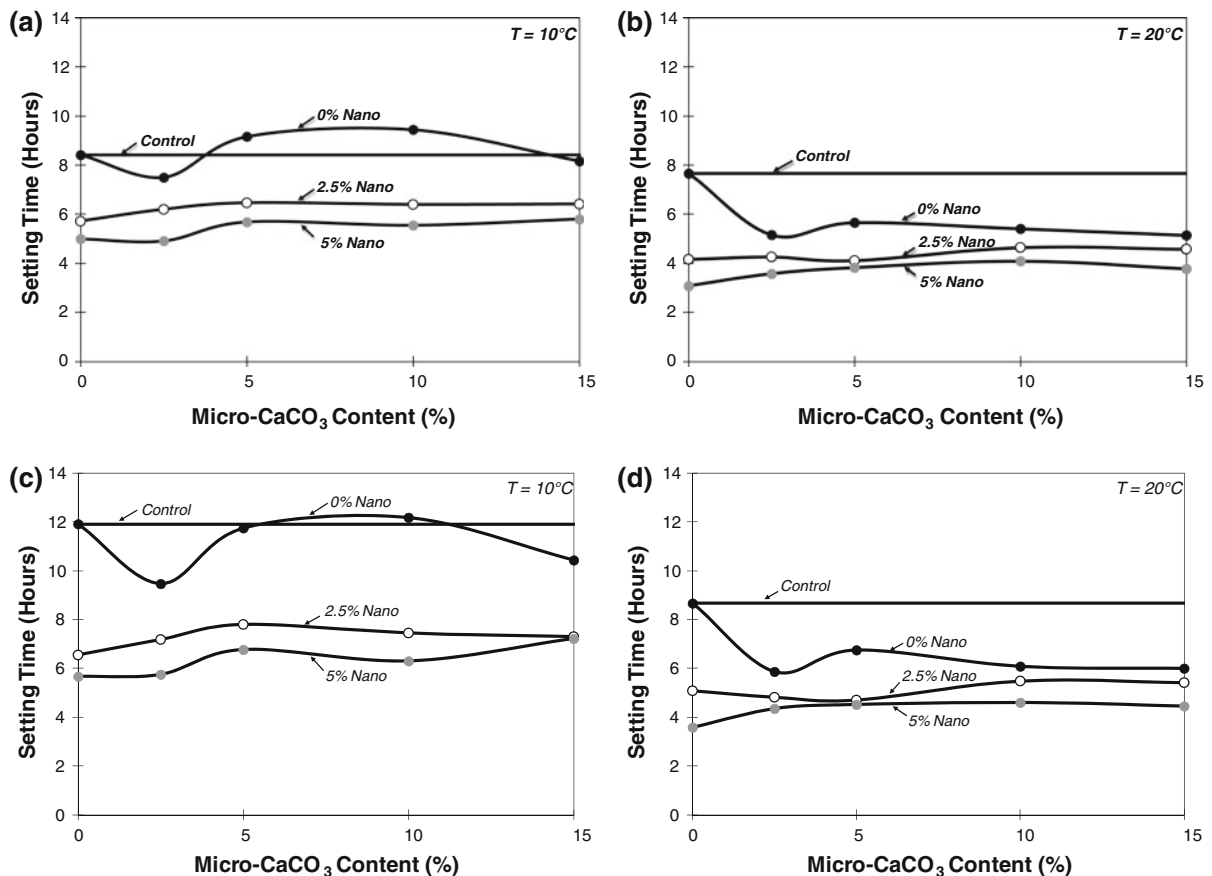


Fig. 3 Setting time results for UHPC mixtures incorporating various sizes and dosages of CaCO_3 , **a** Initial [$T = 10^\circ\text{C}$ (50 °F)], **b** Initial [$T = 20^\circ\text{C}$ (68 °F)], **c** Final [$T = 10^\circ\text{C}$ (50 °F)] and **d** Final [$T = 20^\circ\text{C}$ (68 °F)]

added micro- CaCO_3 particles, which densify the microstructure through enhanced particle packing, and consequently reduce the amount of available space for hydration products to form. Furthermore, the micro- CaCO_3 particles act as contact points between hydration products, eventually solidifying the microstructure to result in shorter setting time [35].

Apparently, there are three interrelated factors influencing the setting time of UHPC mixtures incorporating micro/nano- CaCO_3 : filling, dilution and nucleation sites. Initially, for mixtures incorporating micro- CaCO_3 alone, adding 2.5 % micro- CaCO_3 exhibited lower setting time compared to that of the control mixture (Fig. 3). This can be ascribed to the increase in the number of contact points (i.e. filling) and reduction in the inter-particle space that needs to be filled by hydration products [36]. However, replacing cement with micro- CaCO_3 also induces a cement dilution effect. The filling effect seems to be the dominant at that level of CaCO_3 addition. Increasing the micro- CaCO_3 content (i.e. >2.5 %) accentuates the dilution effect and reduces the rate of hydration products formation, thus leading to a longer setting time. This prolonging in setting time was more pronounced at 10 °C (50 °F) than that at 20 °C (68 °F). At 10 °C (50 °F), the slower rate of hydration due to the dilution effect is boosted by the reduction in the rate of cement hydration reactions induced the low curing temperature as mentioned earlier. Conversely, at 20 °C (68 °F), the hydration acceleration due to the higher curing temperature offsets the retardation induced by the dilution effect.

At higher micro- CaCO_3 contents, the setting time starts to decrease as the micro- CaCO_3 content increases. This indicates that the filling effect starts to dominate the behaviour over the dilution effect. Lower amount of hydration products is needed to fill the inter-particle space, which minimizes the effect of the low hydration reactions rate induced by dilution. This was confirmed via the degree of hydration results as explained later.

On the other hand, adding nano- CaCO_3 induces additional nucleation sites and consequently accelerates the hydration process and offsets the dilution effect. Therefore, mixtures incorporating nano- CaCO_3 exhibited shorter setting time compared to that of mixtures without nano- CaCO_3 . At high micro- CaCO_3 content, increasing the nano- CaCO_3 (i.e. providing more nucleation sites to accelerate the hydration process) may not affect the setting time significantly since at this cement

replacement level the dilution effect is dominant over some increase in the hydration rate.

4.3 Compressive strength

UHPC mixtures are characterized by a very low w/cm ratio; leading to very limited space for cement hydration products to form [6]. Moreover, capillary pores in such concrete typically become discontinuous relatively early; further improvement in mechanical strength depends on the ability of the cementitious materials to hydrate [35]. Therefore, adding nano- CaCO_3 can impart two key effects: (i) It can enhance the packing density of the granular skeleton, and consequently few hydration products are needed to gain strength [37]. As shown in Fig. 4, micro- and nano- CaCO_3 effectively fill voids leading to denser microstructure. Moreover, (ii) these ultra-fine materials fill-up the hollow space formed around agglomerated SF particles; hence, providing a solution for a common problem that occurs in UHPC and affects its strength [38, 39]. Furthermore, CaCO_3 acts as crystallization nucleus for the precipitation of CH [40] leading to an acceleration of the hydration of cement grains. These simultaneous effects result in a higher early strength gain of UHPC.

Experimental results indicate that the curing temperature can significantly affect the early-age compressive strength of UHPC. Figure 5a, b illustrates the 24-h compressive strength results for UHPC mixtures cured at 10 °C (50 °F) and 20 °C (68 °F). Higher curing temperature resulted in higher compressive strength as expected. For instance, the compressive strength at 24-h for mixtures incorporating 0–2.5 % micro- CaCO_3 cured at 10 °C (50 °F) achieved only 40–52 % of that cured at 20 °C (68 °F), respectively.

As shown in Fig. 5a, for mixtures cured at 10 °C (50 °F), adding nano and/or micro CaCO_3 resulted in higher 24-h compressive strength compared to that of the control mixture without CaCO_3 addition, which is in agreement with previous work [41]. For instance, adding 2.5 % micro- CaCO_3 exhibited about 50 % higher 24-h compressive strength compared to that of the control mixture. This improvement in the 24-h compressive strength for mixtures incorporation micro- CaCO_3 with respect to the control mixture decreased as the micro- CaCO_3 content increased. For instance, mixtures incorporating 15 % micro- CaCO_3 exhibited only about 8 % improvement in the 24-h

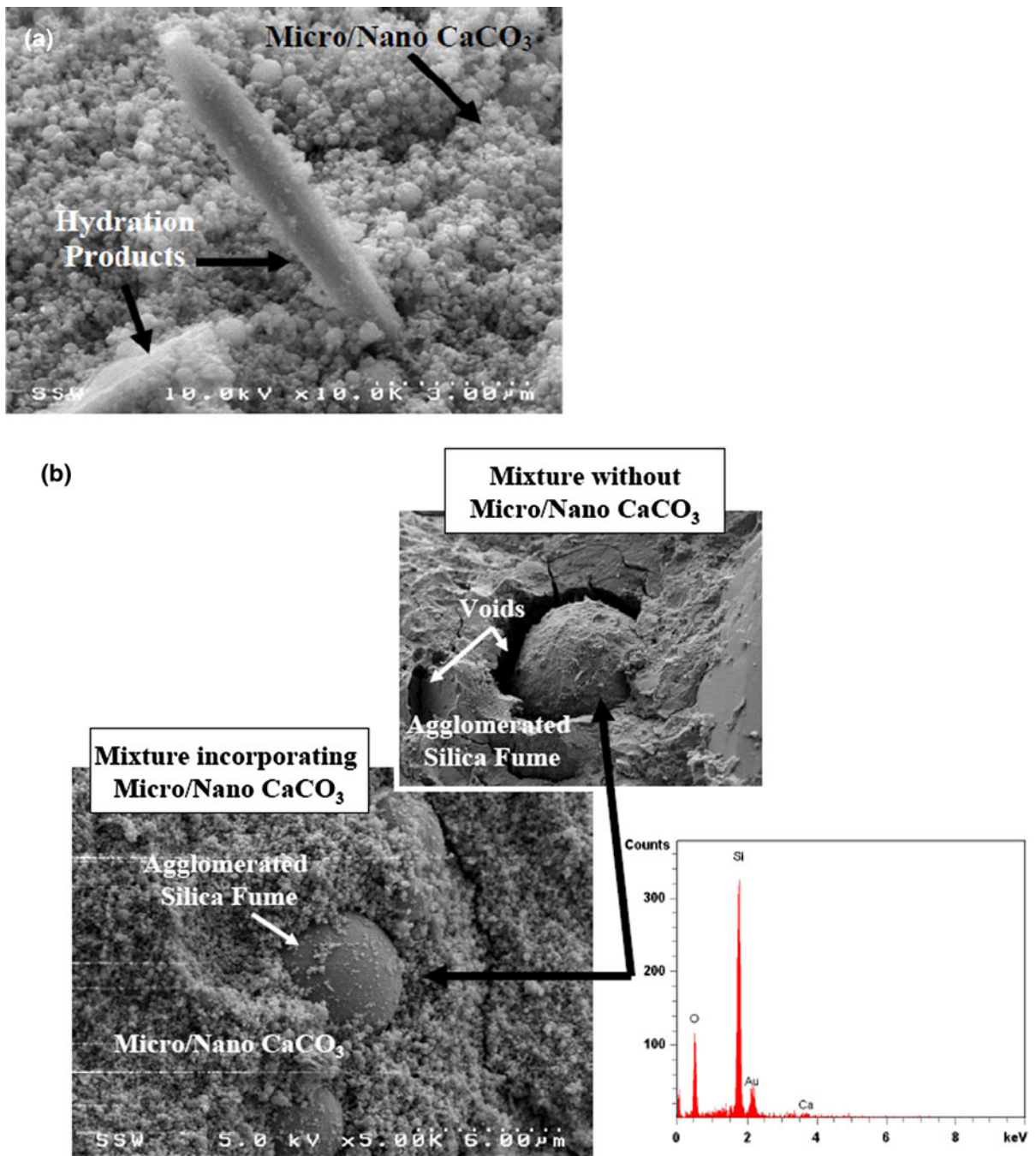


Fig. 4 Micro/nano CaCO_3 **a** covering and filling voids between hydration products and **b** filling hollow space around agglomerated silica fume particles

compressive strength with respect to that of the control mixture. This can be ascribed to the dilution effect induced by replacing a higher amount of cement with CaCO_3 , which acts primarily as an inert filler material.

Similar to micro- CaCO_3 , adding nano- CaCO_3 improved the 24-h compressive strength compared that of to mixtures without nano- CaCO_3 addition. For instance, adding 2.5–5 % nano- CaCO_3 induced about

32–75 % improvement in the 24-h compressive strength with respect to that of the control mixture without CaCO_3 , respectively. This can be attributed to nucleation effect induced nano particles as discussed earlier. Moreover, adding nano- CaCO_3 to mixtures incorporating micro- CaCO_3 also improved the achieved early strength. For instance, adding 2.5 % nano- CaCO_3 to the mixture incorporating 15 % micro- CaCO_3 resulted in about 40 % increase in the 24-h compressive strength.

On the other hand, increasing the nano- CaCO_3 addition beyond 2.5 % did not induce a significant difference (i.e. ± 4 %) in the achieved 24-h compressive strength, regardless of the micro- CaCO_3 content (Fig. 5a). Although the nano-particles accelerate the hydration process through nucleation [2], mixtures with 5 % nano- CaCO_3 may not have sufficient available space for hydration products to form, therefore limiting this accelerating effect. In addition, as the micro- CaCO_3 content increased from 2.5 to 15 %, the 24-h compressive strength for mixtures incorporating a constant level of nano- CaCO_3 did not change significantly (i.e. ± 7 %). This can be considered as a resultant of two compensating effects induced by nano- CaCO_3 addition: acceleration and dilution. During early-age, the nucleation effect induced by nano- CaCO_3 accelerates the hydration process [2]. Simultaneously, adding this nano- CaCO_3 as a replacement for cement increases the dilution effect, thus leading to lower cement hydration progress.

At 20 °C (68 °F), similar to mixtures cured at 10 °C (50 °F), increasing the added amount of nano- CaCO_3 in mixtures from 2.5 to 5 % did not show a significant effect on the 24-h compressive strength. For instance,

increasing the added amount of nano- CaCO_3 from 2.5 to 5 % resulted in a variation in the 24-h compressive strength ranging between -0.5 and $+9.8$ % for mixtures incorporating 0–15 % micro- CaCO_3 . Moreover, mixtures incorporating nano- CaCO_3 exhibited similar 24-h compressive strength to that of mixtures with only micro- CaCO_3 . This can be explained as follows: the development rate of early age strength is strongly linked to the progress of hydration reactions. As mentioned earlier, the limited space for hydration products to form is the main obstacle in UHPC. At a low curing temperature [i.e. 10 °C (50 °F)], the relatively slow rate of hydration reactions is a dominant factor; hence, it longer time is needed to form enough hydration product to fill the limited available space. Therefore, the main source for early strength gain at that low curing temperature is the hydration acceleration induced by nano- CaCO_3 addition, which offsets the low rate of hydration as mentioned previously. Conversely, at higher curing temperature [i.e. 20 °C (68 °F)], the main hurdle is not the rate of hydration but the available space for precipitation and growth of hydration products [42]. Therefore, the nano- CaCO_3 hydration acceleration effect will have a limited influence, while particle packing density will play a major role in compensating for the dilution effect [43].

Figure 6a, b illustrates the 28-days compressive strength results for mixtures cured at 10 °C (50 °F) and 20 °C (68 °F). At 10 °C (50 °F), mixtures incorporating only nano- CaCO_3 exhibited ~ 15 –25 % higher compressive strength compared to that of the control mixture without CaCO_3 . However, adding

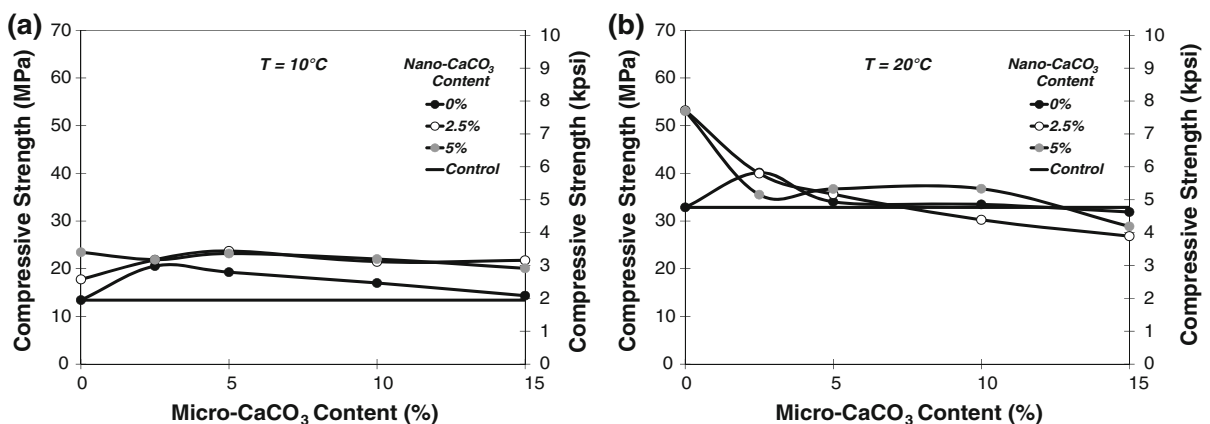


Fig. 5 Twenty-four hours compressive strength results for UHPC mixtures incorporating various sizes and dosages of CaCO_3 cured at **a** 10 °C (50 °F) and **b** 20 °C (68 °F)

nano- CaCO_3 to mixtures incorporating micro- CaCO_3 did not have a significant effect on the compressive strength. Moreover, all mixtures incorporating nano- and/or micro- CaCO_3 showed variation in the achieved 28-days compressive strength of $\pm 6\%$ with respect to that of the control mixture. In this case, the dilution effect was compensated for by the packing effect induced by adding the fine CaCO_3 , thus maintaining comparable compressive strength.

At a curing temperature of 20°C (68°F), all mixtures had similar ($\pm 10\%$) 28-days compressive strength with respect to that of the control mixture. Adding nano- CaCO_3 slightly lowered the compressive strength due to the dilution effect. Furthermore, adding up to 2.5% micro- CaCO_3 appeared to increase the 28-days compressive strength, which can be attributed to the filler effect. With greater additions of micro- CaCO_3 , a decrease in the compressive strength occurred as a result of dilution.

The effect of dilution on the 24-h compressive strength for mixtures cured at 10°C (50°F) and 20°C (68°F) is shown in Fig. 7a, b. At 10°C (50°F), mixtures incorporating CaCO_3 achieved similar to or higher compressive strength than that of the control mixture without CaCO_3 . The highest compressive strength was achieved by mixtures incorporating nano- CaCO_3 , more specifically mixtures incorporating 2.5% nano- and 5% micro- CaCO_3 , and mixtures with 5% nano- and 0 , or 5% micro- CaCO_3 . For these mixtures, between 5 and 10% of cement can be replaced with CaCO_3 , while achieving ~ 73 – 77%

higher 24-h compressive strength than that of the control mixture. The high compressive strength was likely induced by the nucleation effect caused by the presence of nano- CaCO_3 [2]. Above 10% cement replacement, there is only a slight reduction in compressive strength with increasing additions of CaCO_3 . This is because there is not sufficient space in the microstructure to incorporate both filler packing materials [35] and increased amounts of hydration products, which combined, can compensate for the dilution effect.

At 20°C (68°F), the highest 24-h compressive strength was achieved in mixtures incorporating either 2.5 – 5% nano- CaCO_3 , and no micro- CaCO_3 . These mixtures achieved 62% higher 24-h compressive strength than that of the control mixture. Thus, 2.5 up to 5% of cement can be replaced by CaCO_3 while achieving higher early-age compressive strength. In these mixtures, the particle packing density effect is dominant, which reduces the amount of available space in the microstructure [35], therefore limiting the effect of nucleation. However, dilution governs for mixtures incorporating more than 5% CaCO_3 , and the compressive strength tends to decrease beyond a 5% replacement rate.

The effect of the total cement replacement on the 28-days compressive strength is illustrated in Fig. 8a, b. The same early-age trend with respect to the control mixture can be found at later age for the respective curing temperature. At 10°C (50°F), the highest 28-days compressive strength (21% above that of the control mixture) was achieved by the mixture

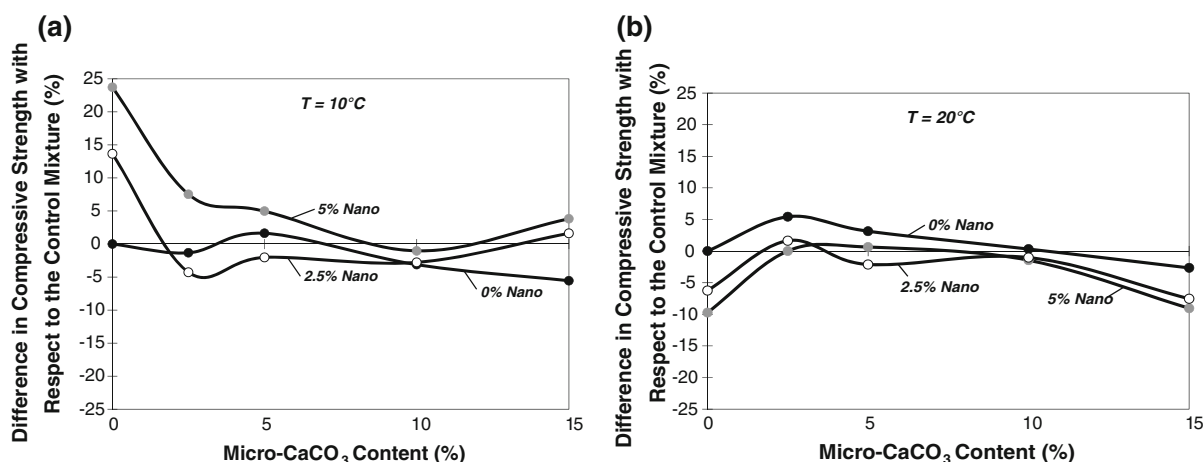


Fig. 6 The 28-days compressive strength results for UHPC mixtures incorporating various sizes and dosages of CaCO_3 cured at **a** 10°C (50°F) and **b** 20°C (68°F)



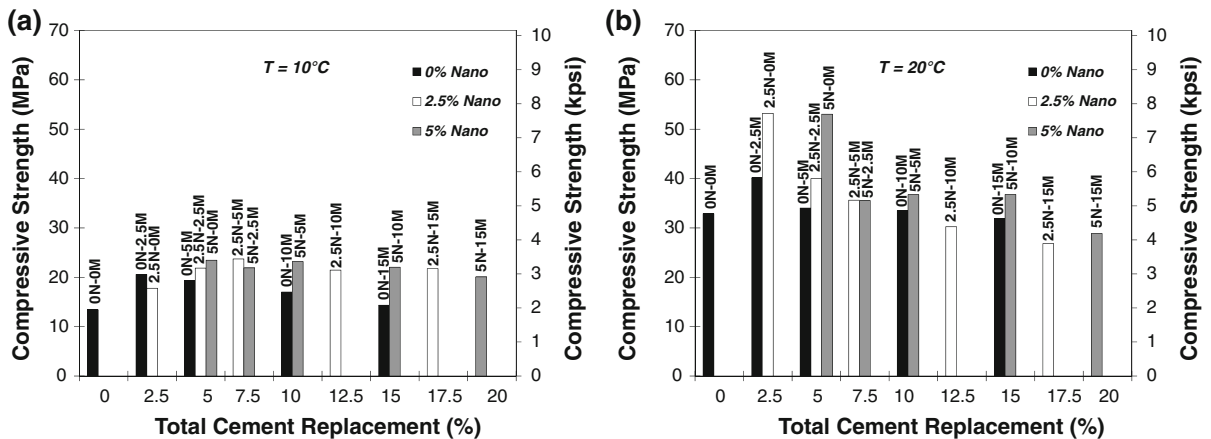


Fig. 7 Twenty-four hours compressive strength results of UHPC mixtures incorporating CaCO₃ in terms of partial cement replacement, at **a** 10 °C (50 °F), and **b** 20 °C (68 °F)

incorporating 5 % nano- and 0 % micro-CaCO₃. At 20 °C (68 °F), the highest 28 days compressive strength (5.4 % above that of the control) was achieved by the mixture incorporating 0 % nano- and 2.5 % micro-CaCO₃. Above this level, dilution governs and was not compensated for by the improved packing density induced by both the nano- and micro-CaCO₃ particles.

4.4 Degree of hydration

The degree of hydration results measured using TGA are summarized in Figs. 9 and 10 for selected mixtures incorporating 15 % micro-CaCO₃ and 5 % nano-CaCO₃, respectively. Similar trend was found for the

other mixtures. It is expected that the degree of hydration at 10 °C (50 °F) would be lower than that at 20 °C (68 °F) due to the lower rate of hydration reactions [42]. However, there are two important observations from the figures. First, a higher degree of hydration was achieved in mixtures incorporating nano-CaCO₃ with respect to that of the control and micro-CaCO₃ mixtures at 10 °C (50 °F) compared to that at 20 °C (68 °F). This indicates the effectiveness of the acceleration effect induced by nano-CaCO₃ at lower curing temperature and its ability to offset the slower rate of hydration and overcome the cement dilution effect. This is in agreement with the compressive strength results discussed earlier.

Second, the variation in the degree of hydration with respect to the variation in the amount of added

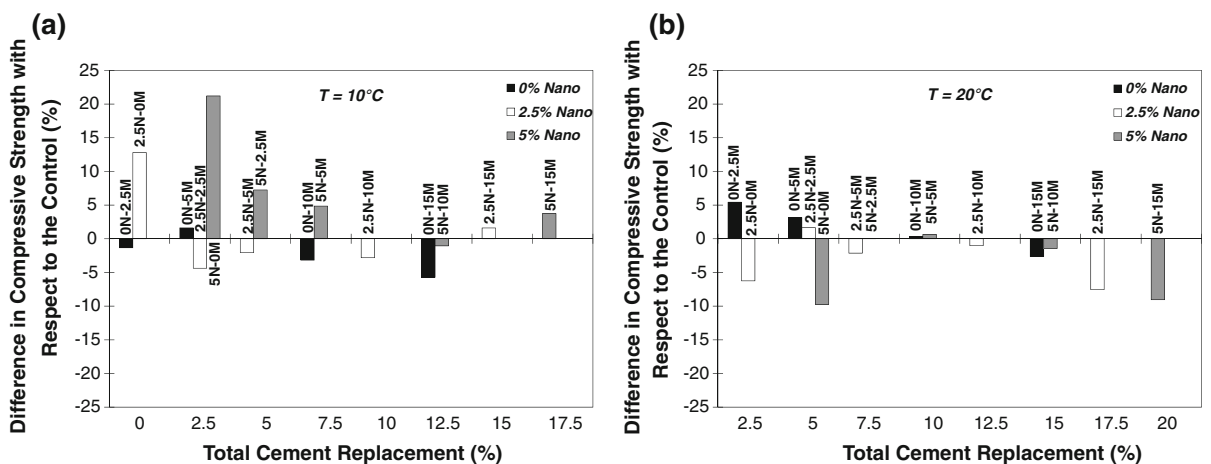


Fig. 8 Twenty-eight days compressive strength results of UHPC mixtures incorporating CaCO₃ in terms of partial cement replacement, at **a** 10 °C (50 °F), and **b** 20 °C (68 °F)

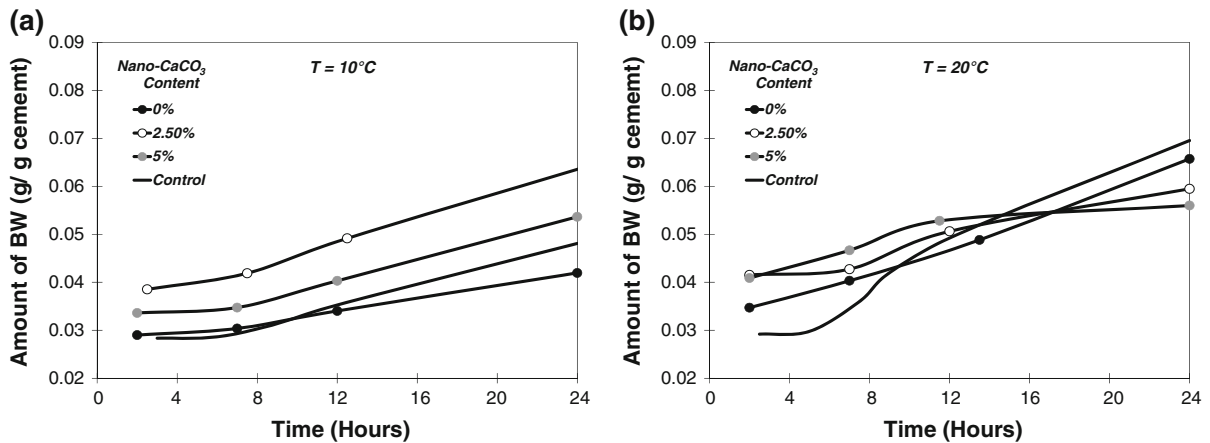


Fig. 9 Degree of hydration results for UHPC mixtures incorporating 15 % micro-CaCO₃ cured at **a** 10 °C (50 °F), and **b** 20 °C (68 °F)

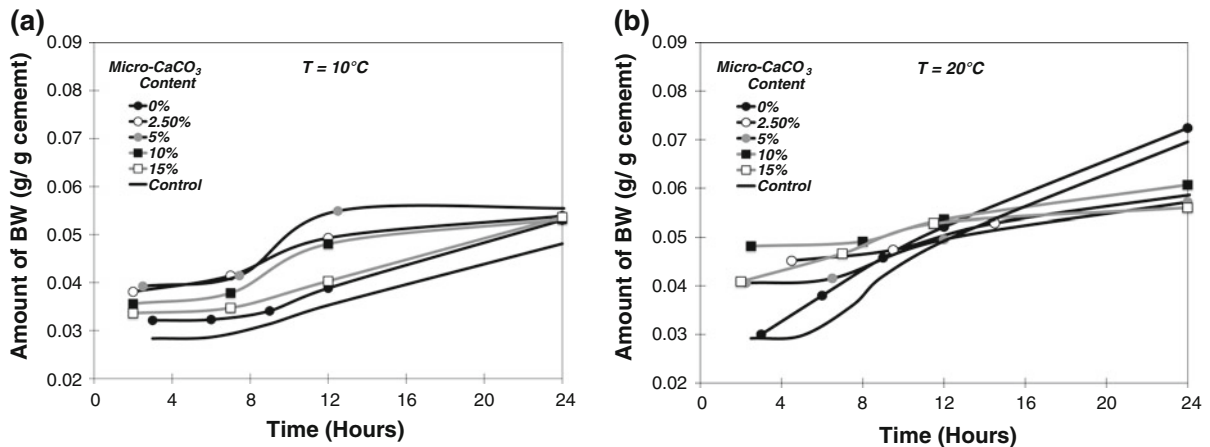


Fig. 10 Degree of hydration results for UHPC mixtures incorporating 5 % nano-CaCO₃ cured at **a** 10 °C (50 °F), and **b** 20 °C (68 °F)

CaCO₃ is more significant at 10 °C (50 °F) compared to that at 20 °C (68 °F). For instance, adding 2.5–5 % nano-CaCO₃ to the mixture incorporating 15 % CaCO₃ (i.e. the total CaCO₃ content is 17.5–20 %, respectively) results in variation in the degree of hydration of about 30–60 % at 10 °C (50 °F) and only 7–13 % at 20 °C (68 °F), respectively. The higher the increase in the degree of hydration, the higher is the amount of hydration products formed. This emphasizes the different effect of nano-CaCO₃ content at different temperatures. However, for the same CaCO₃ content (i.e. the same effect on particle packing density), the available space will be filled relatively slower at lower curing temperature, while adding nano-CaCO₃ accelerates the process significantly

compared to that of mixtures without nano-CaCO₃. At higher temperature [i.e. 20 °C (68 °F)], the hydration process proceeds at higher rate and the available space will be filled by hydration products; hence, the hydration acceleration effect will have a marginal effect. Moreover, the higher the CaCO₃ content, the lower is the available space, and consequently the less significant is the hydration acceleration effect induced by nano-CaCO₃ addition.

In conclusion, at lower curing temperature, the acceleration effect induced by additional nano-CaCO₃ nucleation sites is the dominant mechanism controlling the development rate of hydration reactions and offsetting the cement dilution effect. At higher temperature, the filling and particle packing density

mechanism is dominant, densifying the microstructure and leading to higher compressive strength with relatively lower hydration products formed.

4.5 Heat of hydration

The heat of hydration results for UHPC mixtures containing 5 % nano- CaCO_3 are shown in Fig. 11. The heat of hydration peaks shifted to the left when nano- CaCO_3 was incorporated into mixtures at constant micro- CaCO_3 content, indicating an acceleration of hydration reactions. This is mainly due to the nucleation effect [2], which supports the early-age compressive strength results discussed earlier. For mixtures with a constant nano- CaCO_3 content, incorporating micro- CaCO_3 resulted in temperature peaks that were shifted downwards, and at times slightly to the right, relative to mixtures with only nano- CaCO_3 . This trend signifies that the micro- CaCO_3 particles act as a filling material, and do not contribute significantly to the hydration process [2]. Therefore, the downward shift is apparently due to the dilution effect. The outcome of this trend can be observed in the setting time results since micro- CaCO_3 had little effect, and if any, a slight increase in the setting time compared to mixtures with only nano- CaCO_3 .

4.6 Drying shrinkage

Up to 7-days, generally adding nano- CaCO_3 to UHPC mixtures at constant micro- CaCO_3 cured at 10 °C (50 °F) resulted in decreased shrinkage and similar mass loss compared to that of mixtures with only micro-

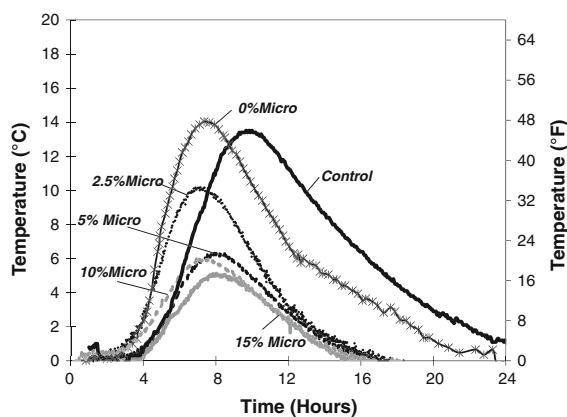


Fig. 11 Heat of hydration results for UHPC mixtures incorporating 5 % nano- CaCO_3 and various dosages of micro- CaCO_3

CaCO_3 . For instance at 7 days, in mixtures with 15 % micro- CaCO_3 , increasing the nano- CaCO_3 content from 0 to 5 % resulted in 18 % lower shrinkage and 3 % lower mass loss (Fig. 12). The presence of ultrafine CaCO_3 particles releases entrapped water [27, 28], which is then incorporated into the hydration process to produce more hydration products. At that low level of w/c, this free water increases the internal relative humidity, leading to lower capillary stresses according to Kelvin's law [44], and consequently reducing the amount of autogenous shrinkage [45]. Furthermore, the nano-particles accelerate the hydration process through inducing the nucleation effect, leading to faster development of internal microstructure, which acts as an internal restraint against shrinkage [46]. Therefore, the reduction in shrinkage strains can be attributed to the lower autogenous shrinkage and the presence of an internal restraint mechanism. Although higher amounts of free water are expected in mixtures with larger quantities of fine particles, the mass loss remained similar due to water being incorporated into the hydration products. In addition, the accelerated hydration process allows for faster depercolation of capillary pores compared to that of mixtures with only micro- CaCO_3 , which lowers the amount of evaporable water.

At 20 °C (68 °F), adding nano- CaCO_3 to UHPC mixtures incorporating micro- CaCO_3 resulted in higher shrinkage strains compared to that of mixtures with only micro- CaCO_3 . Figure 13 illustrates the drying shrinkage and mass loss for mixtures incorporating 15 % micro- CaCO_3 cured at 20 °C (68 °F). At 7-days, mixtures with 15 % micro- CaCO_3 incorporating either 2.5 or 5 % nano- CaCO_3 exhibited 27 and 13 % higher shrinkage, respectively, than that of mixtures without nano- CaCO_3 . At the warmer temperature of 20 °C (68 °F), mixtures containing up to 2.5 % nano- CaCO_3 experienced slightly accelerated hydration induced by the nucleation effect, resulting in higher amounts of autogenous shrinkage. Furthermore, mixtures containing more than 2.5 % nano- CaCO_3 had reduced amount of space for hydration products to form, limiting the accelerating effect, and causing the remainder of the nano- CaCO_3 to act as an inert filler, restraining further shrinkage by creating a denser microstructure. Mixtures with 15 % micro- CaCO_3 experienced a slight, if any, increase in the mass loss due to slightly more free water evaporating [46], since less water has been combined into the hydration process due to the limited space in the microstructure.

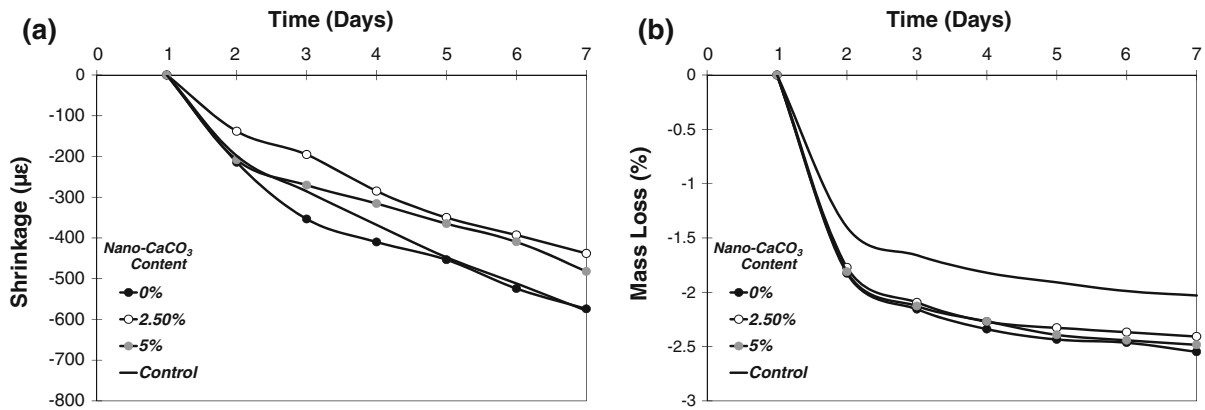


Fig. 12 **a** Drying shrinkage, and **b** mass loss results for UHPC mixtures incorporating 15 % micro-CaCO₃, cured at 10 °C (50 °F)

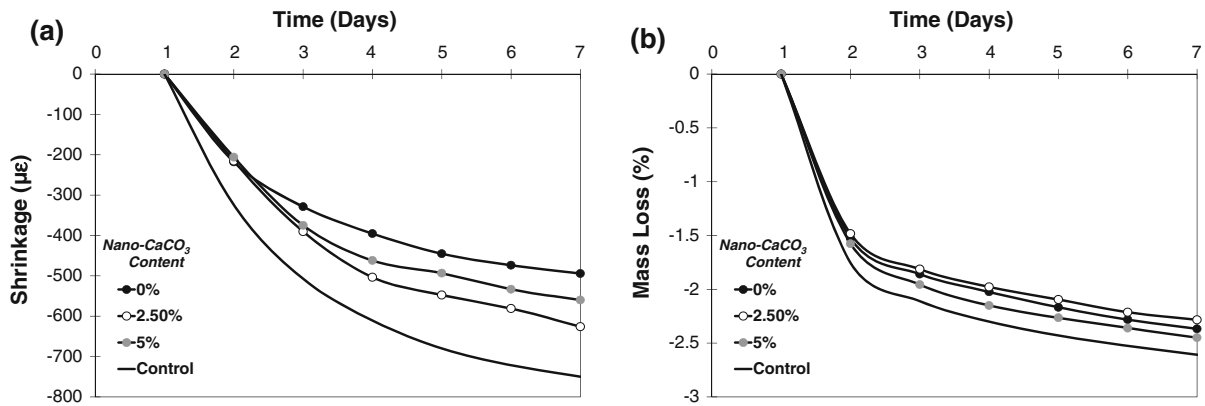


Fig. 13 **a** Drying shrinkage, and **b** mass loss results for UHPC mixtures incorporating 15 % micro-CaCO₃, cured at 20 °C (68 °F)

Mixtures incorporating various dosages of micro-CaCO₃ with a constant nano-CaCO₃ content cured at 10 °C (50 °F) had reduced shrinkage strains and generally higher mass loss compared to that of mixtures with only nano-CaCO₃ (Fig. 14). For instance at 7-days, the shrinkage decreased by 23 %, and the mass loss increased by 6.6 % for the mixture with 2.5 % nano- and 15 % micro-CaCO₃ compared to that of the mixture containing only 2.5 % nano-CaCO₃. The addition of micro-CaCO₃ released some entrapped water, leading to higher evaporation levels as indicated by the results. Higher dilution levels can account for the lower shrinkage rates [47], as less autogenous shrinkage is expected.

At a curing temperature of 20 °C (68 °F), adding micro-CaCO₃ to mixtures incorporating a constant level of nano-CaCO₃ resulted in increased shrinkage and higher mass loss with respect to the mixtures with

only nano-CaCO₃ (Fig. 15). Furthermore, increasing the amount of micro-CaCO₃ from 2.5 to 15 % resulted in no change in shrinkage, but greater mass loss. For instance at 7 days, mixtures incorporating 2.5 % nano-CaCO₃ resulted in 4 % lower shrinkage and 15 % higher mass loss when increasing the amount of micro-CaCO₃ in the mixture from 5 to 15 %. With greater quantities of fine particles, more entrapped water is released [27, 28]. Since micro-CaCO₃ does not affect the hydration process, higher evaporation results are expected, which leads to higher drying shrinkage [46].

Seven days drying shrinkage and mass loss results for mixtures cured at 10 °C (50 °F) are presented in Fig. 16. All mixtures incorporating CaCO₃ exhibited similar or lower shrinkage strains compared to that of the control mixture without CaCO₃. The smallest amount of shrinkage was found in the mixture

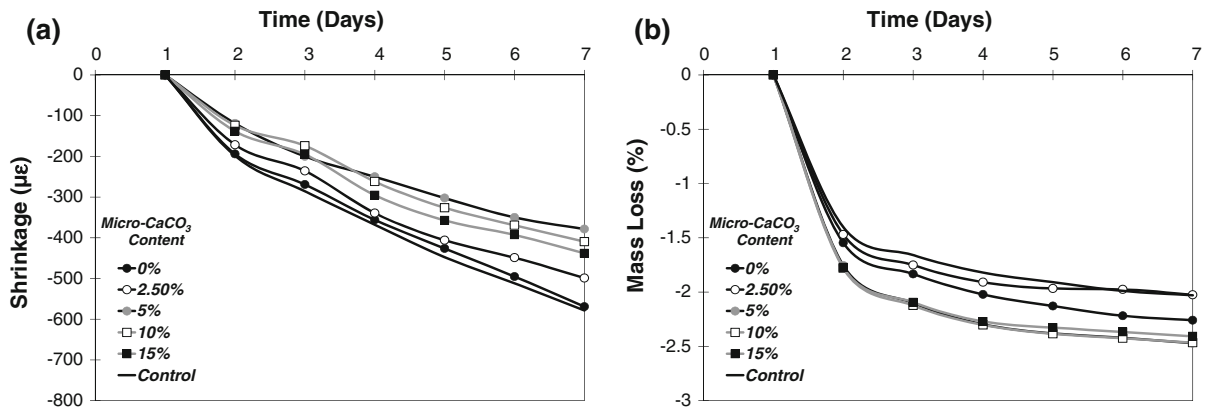


Fig. 14 **a** Drying shrinkage, and **b** mass loss results for UHPC mixtures incorporating 2.5 % nano-CaCO₃, cured at 10 °C (50 °F)

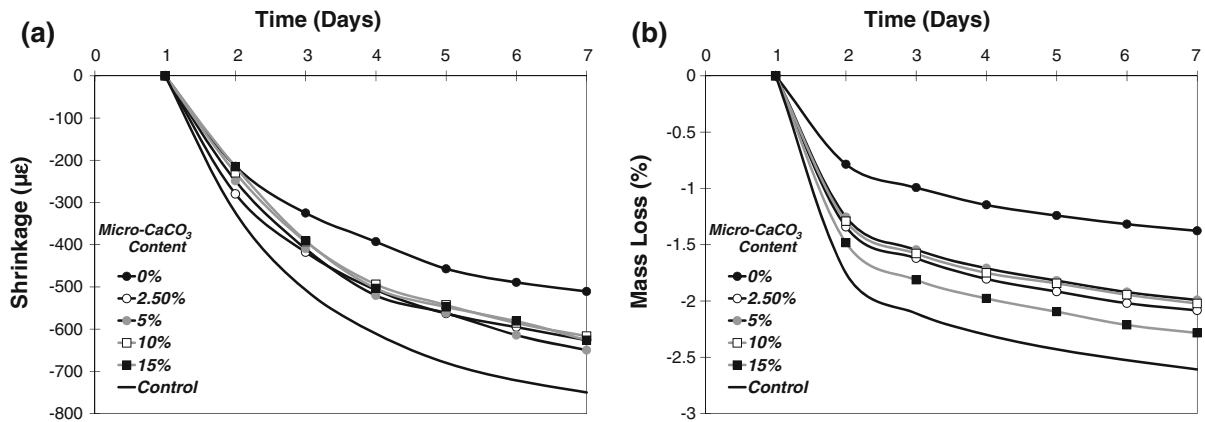


Fig. 15 **a** Drying shrinkage, and **b** mass loss results for UHPC mixtures incorporating 2.5 % nano-CaCO₃, cured at 20 °C (68 °F)

incorporating 2.5 % nano- and 5 % micro-CaCO₃. For this mixture, the reduction in shrinkage was 35 % compared to that of the control mixture. Above this cement replacement value, higher shrinkage rates can be attributed to more free water available for evaporation, therefore leading to higher mass loss.

Figure 17 illustrates the effect of the total cement replacement on the 7 days drying shrinkage and mass loss of mixtures cured at 20 °C (68 °F). Similar to mixtures cured at 10 °C (50 °F), all mixtures incorporating CaCO₃ exhibited less shrinkage compared to that of the control mixture. Three mixtures tied for the smallest amount of shrinkage: 2.5 % nano- and 0 % micro-CaCO₃, 0 % nano- and 10 % micro-CaCO₃, and 0 % nano-15 % micro-CaCO₃, with ~31–34 % decrease in shrinkage compared to that of the control

mixture. Apparently, it is possible to replace up to 15 % of cement with micro-CaCO₃, and achieve similar shrinkage results to that when replacing 2.5 % of cement with nano-CaCO₃.

5 Conclusions

This study investigated the effects of incorporating micro- and nano-CaCO₃, both individually and combined, as partial replacement for cement on the early-age properties and compressive strength of UHPC. Mixtures were tested at both simulated cold [10 °C (50 °F)] and normal [20 °C (68 °F)] conditions. From the experiments conducted with the materials used in this study, it can be concluded that:

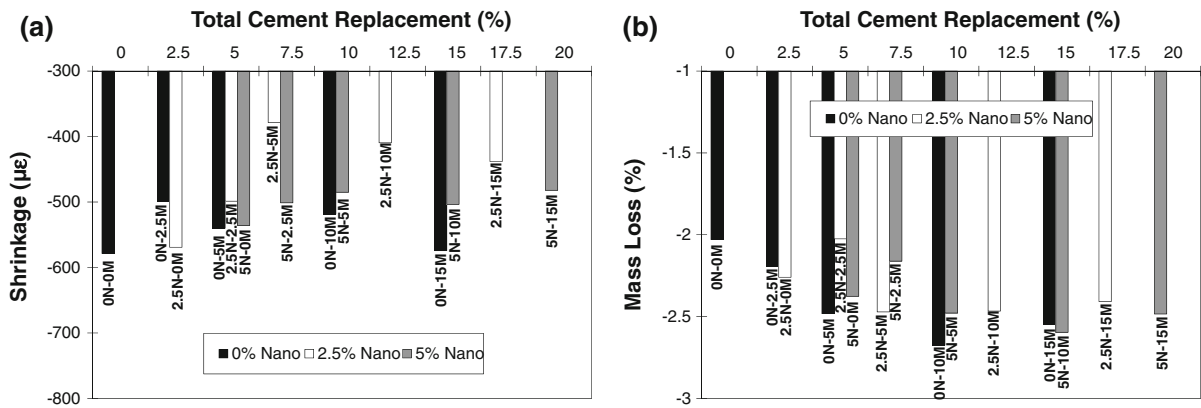


Fig. 16 Results for 7 days **a** drying shrinkage, and **b** mass loss, in terms of partial cement replacement, cured at 10 °C (50 °F)

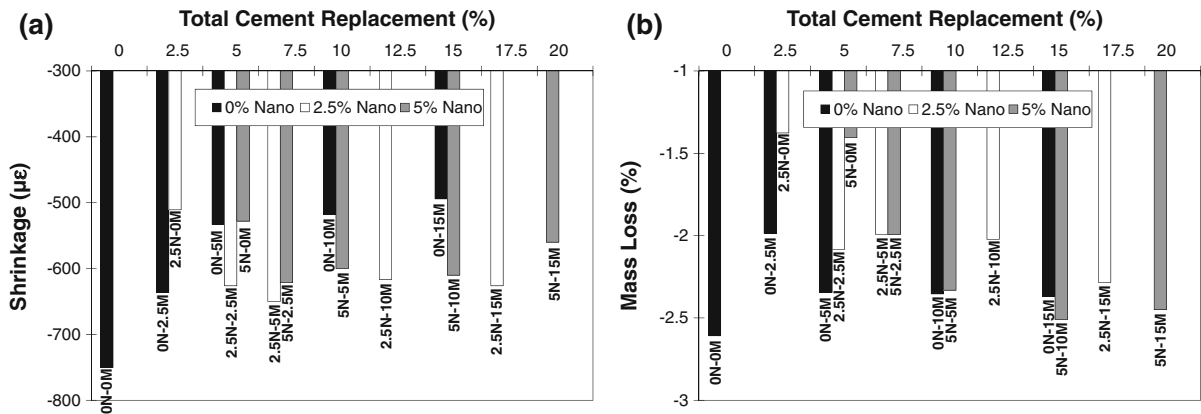


Fig. 17 Results for 7 days **a** drying shrinkage, and **b** mass loss, in terms of partial cement replacement, cured at 20 °C (68 °F)

1. Incorporating micro and/or nano- CaCO_3 resulted in higher flowability of UHPC mixtures compared to that of the control mixture. Increasing the cement replacement level led to more flowable mixtures.
2. Incorporating both nano- and micro- CaCO_3 into cement reduced the setting time significantly, regardless of the curing temperature. Mixtures consisting of 5 % nano- CaCO_3 and 0–15 % micro- CaCO_3 achieved the shortest setting times, at both 10 °C (50 °F) and 20 °C (68 °F).
3. At 10 °C (50 °F), densifying the microstructure through enhanced particle packing density, and the more rapid growth of hydration products induced by nano- CaCO_3 through additional nucleation sites were the main contributors to the improved early-age compressive strength. The highest 24-h compressive strength was achieved by partially replacing cement with 2.5 % nano- and 5 % micro- CaCO_3 , and 5 % nano- and 0 %, or 5 % micro- CaCO_3 . The highest 28-days compressive strength occurred in the mixture incorporating 5 % nano- and 0 % micro- CaCO_3 .
4. At 20 °C (68 °F), densifying the microstructure through enhanced particle packing density was the main contributor to the comparable to or higher early-age compressive strength than that of the control mixture. The lack of available space for hydration products to form limited the accelerated hydration induced by the presence of nano- CaCO_3 . The highest 24-h compressive strength was achieved by replacing cement with 2.5 or 5 % nano- and 0 % micro- CaCO_3 . The highest 28-days compressive strength was achieved in the mixture incorporating 0 % nano- and 2.5 % micro- CaCO_3 .
5. At 10 °C (50 °F), partially replacing cement with 2.5 % nano- and 5 % micro- CaCO_3 achieved the

smallest amount of shrinkage at 7-days. At 20 °C (68 °F), the smallest shrinkage strains were measured in mixtures incorporating 2.5 % nano- and 0 % micro- CaCO_3 , and 0 % nano- and 10 or 15 % micro- CaCO_3 .

6. Through the combination of nano- and micro- CaCO_3 , cementitious materials can have the benefits of accelerated early-age hydration and enhanced particle-packing density, while achieving similar, if not enhanced mechanical properties compared to that of the control mixture without CaCO_3 .

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